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Social Media Conversation Analysis: Sentiment and Toxicity

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CERTIFICATE

This is to certify that the work contained in this report entitled **“Social Media Conversation Analysis: Sentiment and Toxicity”** is submitted by the group members Tushar Pal (Roll No.: 21CSE1036) and Vishal Pratap (Roll No.: 21CSE1040) to the Department of CSE , National Institute of Technology Goa, for the partial fulfillment of the requirements for the degree of **Bachelor of Technology** in **Computer Science and Engineering**.

They have carried out their work under my supervision. This work has not been submitted else-where for the award of any other degree or diploma.

The project work in our opinion, has reached the standard fulfilling of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering in accordance with the regulations of the Institute.

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Tushar Pal
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Abstract

The explosive growth of online social platforms has transformed the landscape of public discourse, but it has also amplified the spread of toxic, hateful, and offensive language. This report presents a comprehensive framework for the detection and analysis of sentiment and toxicity in Reddit conversations, with a focus on discussions surrounding the 2025 Delhi Assembly Elections. We introduce a hybrid deep learning model that combines Bidirectional Long Short-Term Memory (BiLSTM) networks with a Markov probabilistic layer, capturing both global contextual semantics and local sequential dependencies. The architecture is further enhanced with an attention mechanism and supervised attention alignment using the HateXplain dataset, improving interpretability and transparency. To complement our approach, transformer-based models—RoBERTa for sentiment analysis and Toxic-BERT for toxicity detection—are incorporated for nuanced classification. Our preprocessing pipeline addresses the challenges of code-mixed Hinglish text and long-form comments through translation and sliding window inference. Extensive experiments on both benchmark (HateXplain) and custom Reddit datasets demonstrate that the hybrid model achieves a Macro F1-score of 0.6514, outperforming standard BiLSTM baselines while maintaining computational efficiency. Thread reconstruction using directed graphs enables visualization of toxicity propagation within conversations. The results highlight the model's ability to balance accuracy, explainability, and scalability, offering actionable insights for content moderation and public opinion mining in complex, multilingual social media environments.

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Chapter 1

Introduction

The growth of social media has opened up powerful new ways for people to share their thoughts and opinions. However, it has also led to the unchecked spread of toxic, hateful, and offensive content. Platforms like Reddit—where users often interact anonymously in threaded discussions—tend to be especially vulnerable to such harmful communication, particularly in conversations around polarizing topics like politics or identity. Detecting and managing this content is essential to maintaining healthy and respectful online environments.

Conventional techniques for identifying hate speech typically rely either on surface-level text patterns or on complex models that are difficult to scale. While transformer-based models like BERT have demonstrated impressive accuracy, they often come with trade-offs—such as significant memory consumption and slower inference speeds—which limit their practical use in real-time moderation systems [2]. Additionally, many existing solutions do not offer much transparency, making it hard for users or moderators to understand how decisions are being made.

To overcome these limitations, we introduce a lightweight yet powerful hybrid model that combines Bidirectional LSTMs with a probabilistic Markov layer. This setup is designed to capture both long-range context and short-range token transitions, helping the model grasp the structure and tone of a sentence more effectively. We further enhance this system by incorporating an attention mechanism, and when available, we guide the model using rationale-based annotations from the HateXplain dataset [1], which adds interpretability to its predictions.

To improve classification accuracy on emotionally charged content, we complement our model with fine-tuned transformer architectures. For instance, RoBERTa is used for sentiment analysis due to its strength in handling informal and nuanced language [3], while Toxic-BERT is utilized for identifying harmful or abusive content [4].

We also tackle the unique challenge of code-switched language—specifically Hinglish—by building a preprocessing pipeline that standardizes slang, corrects informal expressions, and cleans noisy text using regular expressions. For Reddit's long and complex comments, we apply a sliding window strategy to ensure complete analysis despite token length limitations.

To gain deeper insights into how harmful language travels through conversations, we reconstruct Reddit threads into directed graphs that reflect their reply structures. This allows us to trace the development of sentiment and toxicity over time and across interactions.

Overall, this work presents a modular, explainable, and computationally efficient framework for identifying and analyzing offensive content in social media discussions—particularly in the dynamic and often unpredictable environment of Reddit.

Chapter 2

Related Work

The detection of sentiment and toxicity in social media conversations has emerged as a critical area of research, given its implications for online community health, user experience, and the broader societal discourse[5]. Traditional approaches to analyzing online conversations often relied on manual annotation and rule-based systems, which were limited in scalability and struggled to capture the nuanced expressions of sentiment and toxicity[6]. This has driven the adoption of advanced machine learning and deep learning techniques to enable more accurate, robust, and context-aware analysis of social media interactions.

Early studies in this domain primarily focused on basic sentiment classification, utilizing feature engineering methods such as Bag-of-Words (BOW) and TF-IDF to represent textual data.[6] For example, Hitesh et al. employed Word2Vec embeddings in conjunction with a Random Forest classifier to perform real-time sentiment analysis of Twitter data during the 2019 Indian elections, demonstrating that word embeddings significantly enhance the detection of positive and negative sentiments compared to traditional approaches [7].

As the field progressed, researchers began to address the challenge of detecting and understanding toxicity—a more complex and context-dependent phenomenon than sentiment [8][9]. Aleksandric et al. [5] conducted a large-scale analysis of Twitter conversations to investigate how exposure to toxic comments influences user behavior and emotional responses. Their findings indicated that toxicity not only escalates negative emotions such as sadness and anger but also increases the likelihood of retaliatory or disengaging behaviors among users. By incorporating confounding factors like account identifiability and conversation structure, the study highlighted the cyclical and self-perpetuating nature of toxic interactions, and demonstrated the feasibility of predictive modeling for proactive moderation [5].

With the advent of transformer-based models, the detection of toxic behavior has become increasingly sophisticated. Kumar et al. [10] leveraged BERT and HateBERT—the latter being specifically fine-tuned for hate speech detection—to identify and analyze toxic content in Reddit discussions related to e-commerce and gaming. Their results showed that domain-adapted models like HateBERT outperform general-purpose language models, underscoring the importance of context-specific adaptation for nuanced toxicity detection [10].

Recent research has also explored the propagation and impact of toxicity within the complex structures of online conversations. Shankaran and Sharma [11] studied Reddit threads, revealing that the presence of early toxic comments significantly increases the probability of subsequent toxic replies. Their findings suggest that immediate conversational context is often more influential than the original post in shaping user responses, and that both highly toxic and non-toxic posts can sustain engagement, producing a polarizing effect on participation [11].

Further, works by Falade et al. [12] and Yousefi et al. [13] have advanced the field by focusing on the prediction of toxic comments and examining the broader impact of toxicity on the flow and structure of Reddit conversations. These studies highlight the persistent challenge of toxicity across diverse topics and languages and emphasize the need for scalable, adaptable detection systems that can operate effectively in real-world, multilingual, and multi-domain environments [12][13].

In summary, the literature shows a shift from simple sentiment analysis to advanced, context-aware models for toxicity detection. Leveraging modern machine learning, researchers can now better understand and address the complexities of sentiment and toxicity in social media, promoting healthier online communities.

Chapter 3

Reddit-Based Analysis of Sentiment and Toxicity

This chapter presents an in-depth analysis of sentiment and toxicity within Reddit discussions, emphasizing the distinctive conversational dynamics shaped by Reddit's pseudonymous and community-centric environment. Unlike platforms where user identity is more transparent, Reddit allows individuals to engage in discourse using pseudonyms, which can influence the openness, candor, and sometimes the extremity of expressed opinions [8][5]. This quality makes Reddit a compelling platform for studying the propagation and manifestation of sentiment and toxic behavior in online communities [9][13].

Reddit's architecture is organized around thousands of topic-specific subreddits, each functioning as a semi-autonomous community with its own norms, moderation policies, and user demographics [13][8]. This structure enables researchers to explore sentiment and toxicity not only at a platform-wide level but also within focused thematic contexts, such as political discussions, social issues, or local events [11][10]. For this study, particular attention is given to subreddits that are active in Indian socio-political discourse, including those that engage with topics surrounding the 2025 Delhi Assembly Elections.

The platform's threaded conversation format further enriches the analysis, as it allows for the examination of how sentiment and toxicity evolve within the context of replies, comment chains, and branching discussions [5][12]. This structure provides valuable insights into the dynamics of online argumentation, escalation of toxic exchanges, and the emergence of echo chambers or polarized subgroups [13].

Moreover, Reddit's user base is known for its diversity, both in terms of geography and perspectives, which contributes to a wide spectrum of sentiments and opinions. The prevalence of code-switching and the use of Hinglish—Hindi expressed in the Roman alphabet—adds another layer of linguistic complexity, requiring robust preprocessing and analytical techniques to accurately capture the nuances of user expression.

3.1 What is Reddit?

Reddit is a widely used online platform that blends content sharing with community-driven discussions. Registered users can contribute by posting text, links, images, or videos. These posts are sorted into topic-specific communities called **subreddits**, each centered around a particular subject such as news, technology, or entertainment. Users can vote on posts—either upvoting or downvoting—which influences how prominently content appears on the platform.

3.2 How Reddit works

Each subreddit operates under its own moderation team and follows community-specific guidelines. Comments on Reddit are displayed in a nested format, forming a tree-like structure that helps preserve the context of replies and supports more meaningful, in-depth conversations. This layered design distinguishes Reddit from many other social media platforms in terms of how interactions are organized.

Due to its transparent structure and community moderation, Reddit is especially appealing for research. Tools such as `PRAW (Python Reddit API Wrapper)`[14] and `Pushshift` offer efficient ways to gather large datasets of historical posts and comment threads, making it easier to examine communication patterns and user engagement across various topics [14].

3.3 Why Prefer Reddit?

Reddit stands out as a preferred platform for data collection in social media research due to its distinctive features and openness. One of the primary advantages is its transparency and accessibility; Reddit has consistently offered an open, developer-friendly API with generous rate limits, and researchers benefit further from historical data availability through third-party services like Pushshift. Additionally, Reddit's unique tree-like comment structure enables in-depth contextual and discourse analysis, facilitating a more nuanced understanding of interactions and discussions [8]. The platform also serves as a rich ground for political discourse, particularly within subreddits such as `r/India`, `r/Delhi`, and `r/IndiaSpeaks`, which are known for their high levels of civic engagement and relatively unmoderated, organic debates. Moreover, the anonymity afforded by Reddit—where users typically interact using pseudonyms—encourages more candid expressions of opinion. This often results in users sharing sentiments that might be restrained on identity-based platforms like Facebook or LinkedIn, making Reddit especially valuable for studying uninhibited and diverse public opinions.

3.4 Why Not Twitter?

Although Twitter was initially considered for this study due to its prominence in shaping public discourse, several evolving limitations made it less viable for rigorous academic research. One of the major setbacks was the introduction of strict rate limits and monetized API access tiers following organizational changes in 2023. These changes significantly reduced the accessibility of Twitter's data, especially for independent and academic researchers. Furthermore, the previously available free academic access was deprecated, and steep fees were imposed, creating a financial barrier for large-scale data collection. Compounding these issues, access to user-level metadata and historical tweet archives became increasingly restricted, which hindered the ability to conduct longitudinal analyses and study behavioral trends over time. Additionally, modifications in Twitter's content moderation policies contributed to inconsistencies in the quality and reliability of data, particularly concerning toxicity detection. The rise in bot-generated content further complicated the process of obtaining clean, authentic datasets. Given these constraints, the study pivoted to Reddit as a more structured and transparent alternative, where community-driven discussions and accessible data facilitate richer sentiment and toxicity analysis.

Chapter 4

Data Collection

4.1 Dataset – Delhi Assembly Election Conversations

We chose to focus our project on analyzing social media conversations around the Delhi Legislative Assembly Elections because political discourse in India has become increasingly active and polarized online. The 2025 Delhi Assembly Election provided a rich context for exploring public sentiment, engagement, and toxicity in user-generated content. Platforms like Reddit offer a relatively uncensored and candid space for political discussion, making it ideal for mining and analyzing large volumes of user opinions[8].

Using Async PRAW (Python Reddit API Wrapper)[14], we scraped posts and comments from the identified subreddits, focusing on discussions surrounding the Delhi elections by searching the relevant **keywords** associated with the topic. The metadata collected included post titles, selftext, timestamps, number of comments, subreddit details, and author information (pseudonymous usernames) for **post data** and included post id , username, comment text, comment id, parent comment id and timestamp for **comments** to those posts.

4.2 Data Collection and Scope

The dataset for this study was sourced from a selection of Reddit subreddits that actively engage in Indian political discourse. These included `r/delhi`, `r/india`, `r/unitedstatesofindia`, `r/IndiaSpeaks`, `r/indianews`, `r/indiadiscussion`, and `r/AskIndia`. These communities were chosen due to their consistent activity levels, relevance to political discussions, and diverse user engagement. The data collection spanned from December 2024 to January 2025, a period that was deliberately selected to capture a wide range of electoral sentiment. This timeframe encompassed the critical phases of pre-election debates, campaign narratives, and immediate public reactions following official announcements, providing a comprehensive view of the online political climate in the lead-up to the 2025 Indian Assembly Elections.

4.2.1 Dataset Statistics

- **Total posts collected:** 1,598
- **Total comments collected:** 25,617
- **Relevant posts after filtering:** 432
- **Relevant comments after filtering:** 12,414

4.2.2 Filtering and Preprocessing Pipeline

To ensure the relevance and quality of the dataset, a structured multi-stage filtering and preprocessing pipeline was implemented. The initial step involved language detection, where only English-language posts were retained using a language classification model to maintain consistency in textual analysis. Subsequently, a pre-trained zero-shot classification model was applied to assess the relevance of posts to the Delhi Assembly Elections. Each post was evaluated based on its semantic alignment with predefined election-related labels, and those surpassing a certain confidence threshold were marked as relevant. To avoid discarding potentially meaningful content near the classification boundary, a dynamic thresholding strategy was adopted, allowing for flexible inclusion of borderline cases. Finally, a manual review phase was conducted, during which 156 posts that had been initially flagged as irrelevant were re-evaluated and reinstated. Additionally, 998 comments related to the elections were identified and restored, further enhancing the completeness of the dataset.

4.2.3 Insights from Preliminary Data Exploration

Initial exploration of the cleaned dataset revealed several noteworthy patterns. A keyword frequency analysis highlighted frequent mentions of political entities and figures such as AAP, BJP, Kejriwal, Modi, and Rahul Gandhi, suggesting high levels of user engagement with both regional and national political narratives. Among the various subreddits analyzed, `r/delhi` emerged as a major contributor, with 1,406 relevant posts, indicating a strong localized discourse. Other subreddits, including `r/india` and `r/unitedstatesofindia`, also exhibited substantial activity, reflecting broader national interest. Temporal analysis showed a significant increase in user engagement toward the end of January, coinciding with major election-related announcements and heightened media attention. Additionally, thematic analysis using word clouds revealed prominent terms such as "democracy," "corruption," and "governance," which underscore the diversity and depth of political concerns expressed by users [8].

4.2.4 Relevance and Future Utility

This dataset lays the groundwork for deeper tasks like:

- Sentiment analysis on election-related commentary[6]
- Toxicity detection to measure polarization[1]
- User behavior modeling before and after critical events[5]

4.3 Data Filtering Using Zero-Shot Classification

One of the major challenges we faced after scraping data was the presence of noise—posts and comments unrelated to the election. To address this, we implemented a zero-shot classification pipeline using the facebook/bart-large-mnli model from Hugging Face[4]. Each post's title and body were evaluated against candidate labels such as "Delhi Assembly Election" and "Irrelevant." Posts with a relevance score above a 0.7 threshold were retained for further analysis, ensuring that our dataset remained focused on election-related discussions.

This filtering step was vital in maintaining data integrity and optimizing resource use. By removing irrelevant content, we concentrated the analysis on discussions that directly reflected public sentiment surrounding the elections.

4.4 HateXplain as a Benchmark Dataset

HateXplain serves as a critical benchmark for hate speech detection because it provides not only a three-class labeling scheme (*hate speech*, *offensive speech*, and *normal*) but also human rationales and target community annotations. This multifaceted annotation strategy allows researchers to evaluate their models on standard performance metrics (accuracy, macro F1-score, AUROC) while also examining bias and interpretability.

Specifically, models can be compared against ground-truth rationales to assess how closely machine-generated attention aligns with human judgment. Additionally, the fixed nature of HateXplain as a benchmark dataset promotes reproducibility and fair comparisons across different architectures, ensuring that any reported gains in performance or interpretability are truly indicative of improvements rather than artifacts of varying training conditions or data expansions.

Chapter 5

Methodology

5.1 Overview

Our research methodology consists of four main stages: data collection, data filtering using zero-shot classification, model training on the filtered dataset, and benchmarking on the HateXplain dataset with subsequent inferences from the results. In the first stage, we collect raw data from Reddit, focusing on posts related to the 2025 Delhi Assembly Elections. This raw data reflects the complexity and nested structure typical of online political discussions.

Next, we employ zero-shot classification techniques to filter and clean the collected data, ensuring that only relevant posts—those with significant sentiment and toxicity cues—are retained for further analysis. This filtering step is crucial in reducing noise and improving the quality of the training set.

After data filtering, we train our models on the refined dataset. Our work focuses on benchmarking our trained models against the established HateXplain dataset[1]. By doing so, we evaluate model performance in terms of accuracy, macro F1-score, and AUROC, thereby validating our approach and drawing inferences about its effectiveness. The outcomes of this benchmarking phase provide key insights into our models' capabilities, highlighting strengths such as computational efficiency and areas for improvement, such as addressing overfitting. Future work will continue to build upon these foundational stages.

5.2 Model Introduction and Usage

To comprehensively analyze sentiment and toxicity in Reddit discussions surrounding the Delhi Legislative Assembly Elections, we employed a multi-model architecture tailored for both performance and explainability. The primary models used in this project include the **BiLSTM + Markov Hybrid Model**, along with auxiliary transformer-based models such as `cardiffnlp/twitter-roberta-base-sentiment-latest` for sentiment analysis and `unitary/toxic-bert` for toxicity classification. Below, we detail the motivation, architecture, and application of each model in our pipeline.

5.2.1 BiLSTM + Markov Hybrid Model

This section describes the design and implementation of a hybrid model that combines Bidirectional Long Short-Term Memory (BiLSTM) networks with a Markov layer. The overall goal of this model is to take advantage of deep contextual understanding provided by BiLSTM while incorporating the computational efficiency and local dependency modeling of a Markov process. This blend is especially effective for applications such as sentiment analysis and hate speech detection in social media data.

5.2.2 Bidirectional LSTM (BiLSTM)

BiLSTM is an advanced form of recurrent neural networks that processes sequence data in both forward and backward directions. This dual processing allows the model to capture both past and future context, thus providing a richer understanding of sequential data.

5.2.3 How BiLSTM Works

The Bidirectional Long Short-Term Memory (BiLSTM) architecture enhances sequence modeling by capturing contextual dependencies in both forward and backward directions. The process begins with an embedding layer, where the input text is converted into a numerical representation using pre-trained word embeddings, such as GloVe vectors with 300 dimensions. This embedded sequence is then fed into two separate LSTM layers: a forward LSTM that processes the input from the beginning to the end, and a backward LSTM that simultaneously processes it in reverse order. The outputs from both directions are subsequently concatenated, enabling the model to retain richer contextual information compared to a unidirectional LSTM. Optionally, an attention mechanism can be applied to the concatenated outputs, allowing the model to focus selectively on the most relevant parts of the input sequence, thereby improving interpretability and performance in tasks such as classification or sentiment analysis.

This architecture is effective at learning long-term dependencies in sequential data, which is beneficial for many natural language processing tasks.

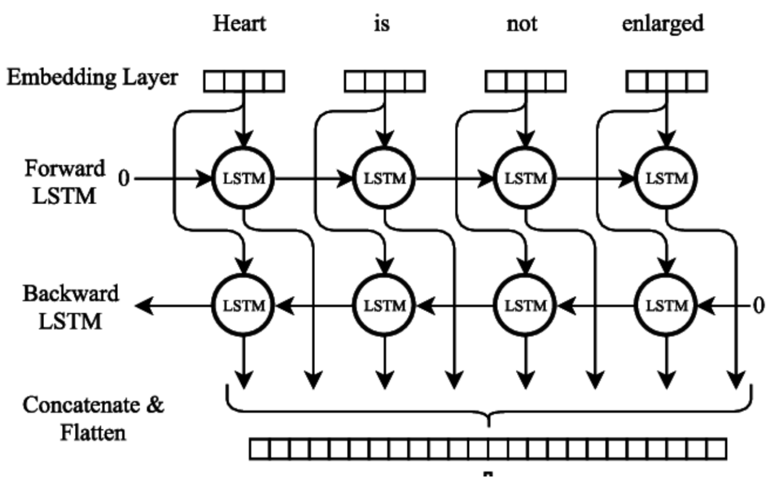

Figure 5.1: Bidirectional Long Short-Term Memory Network.

5.2.3.1 Motivation

The hybrid model was designed to balance computational efficiency with deep contextual understanding. Transformer-based models, while powerful, are often computationally expensive and less suited for real-time or resource-constrained environments. Our BiLSTM + Markov approach offers a lightweight alternative with competitive performance, particularly on structured datasets like HateXplain.[1]

5.2.4 Architecture

The proposed hybrid model combines sequential deep learning and probabilistic modeling to effectively capture both contextual and local linguistic patterns. At the foundation, the embedding layer is initialized with 100-dimensional GloVe vectors[15], which are made trainable during the learning process to adapt to task-specific nuances. This is followed by two stacked bidirectional LSTM (BiLSTM) layers, each comprising 128 hidden units, designed to extract contextual information from both forward and backward directions in the text. An attention mechanism is integrated on top of the BiLSTM outputs to emphasize salient components of the input sequence. When supervised attention is employed—such as using rationale annotations from the HateXplain dataset—it enhances the model's interpretability by aligning attention weights with human judgment [1]. Parallel to this, a Markov branch computes word-to-word transition probabilities, capturing short-range sequential dependencies that may be overlooked by the LSTM layers. Finally, the features from both branches are passed to a classification head, consisting of a dense layer with softmax activation, which predicts whether a given input expresses hate speech, offensive language, or neutral content.

5.2.5 Usage

The developed model was employed to identify instances of hate and offensive speech within Reddit comments, leveraging both contextual understanding and probabilistic patterns. A notable feature of its application was the use of attention alignment guided by human rationales from the HateXplain dataset, which enhanced the model's interpretability by offering insight into its decision-making process. Designed for scalability, the model demonstrated efficiency in handling large volumes of social media data while maintaining a low computational footprint. In terms of performance, it achieved an accuracy of 67% and a Macro F1-score of 0.6514, positioning it competitively among BiRNN-based models. Moreover, it outperformed BERT-based alternatives in terms of speed, offering a practical balance between accuracy and computational cost[1].

5.2.6 Markov Model

A Markov model is a stochastic model that assumes the Markov property, where the future state of a system depends only on its current state and not on the sequence of states that preceded it.

5.2.6.1 How a Markov Model Works

The Markov model operates by defining a set of states along with a transition matrix that quantifies the probability of transitioning from one state to another. In the context of textual data, these states correspond to words or tokens, and the model captures the likelihood of a word following another based on observed frequencies. Its strength lies in modeling local dependencies within sequences, making it particularly effective for identifying token-to-token dynamics. Owing to its reliance solely on the current state to predict the next, the Markov model maintains a low computational footprint, which allows it to scale efficiently even when processing extensive datasets.

5.2.6.2 Rationale for the Hybrid Approach

The integration of a Markov layer into the BiLSTM framework introduces a complementary mechanism that enhances both the efficiency and interpretability of the model. While the BiLSTM captures long-range dependencies and contextual semantics through its deep sequential processing, the Markov component adds a layer of statistical insight by modeling short-range transitions between words. This hybridization enriches the feature space by combining global contextual signals with localized transition patterns. Additionally, the Markov layer offers a lightweight computational component that operates with minimal overhead, making the model more scalable. From an interpretability perspective, the inclusion of transition-based features aids in understanding local token relationships, thereby improving the overall explainability of the model's outputs.

5.2.7 Transformer-Based Sentiment and Toxicity Models

a. Sentiment Classification - RoBERTa

RoBERTa, a transformer-based language model introduced by Facebook AI, serves as a robust evolution of the original BERT architecture. It retains the core architecture of deep transformer encoders with self-attention mechanisms while incorporating several significant enhancements to boost performance in natural language processing tasks. Unlike BERT's static masking, RoBERTa implements dynamic masking, allowing it to randomly mask different tokens during each training iteration. This strategy exposes the model to a wider array of contextual patterns, leading to richer and more resilient language representations. Moreover, RoBERTa discards the Next Sentence Prediction (NSP) objective found in BERT, focusing solely on masked language modeling. This simplification enables better modeling of intra-sentence dependencies, particularly in longer textual sequences. Its training regimen also distinguishes itself by leveraging an extensive corpus exceeding 160GB of text, larger batch sizes, and prolonged training schedules, all of which contribute to more refined contextual understanding. Additionally, RoBERTa employs an expanded Byte-Pair Encoding (BPE) vocabulary, enhancing its ability to manage rare and out-of-vocabulary tokens effectively.

For sentiment classification in this study, we utilized the `cardiffnlp/twitter-roberta-base-sentiment-latest` model, which is specifically optimized for analyzing sentiment in social media contexts. Each Reddit comment was tokenized and processed through a sliding window of 512 tokens with an overlap stride of 256 tokens. The segmented outputs were individually classified, and their results were aggregated using confidence scores and majority voting to determine the final sentiment label—categorized as positive, neutral, or negative. This model was selected due to its demonstrated ability to detect subtle and complex sentiment patterns within informal, user-generated content, which is typical of platforms like Reddit.

RoBERTa functions through a three-stage process. During pre-training, it learns to predict masked tokens in a sequence by relying on the surrounding context, enabling the development of deep bidirectional contextual representations. In the fine-tuning stage, the model adapts to specific downstream tasks, such as sentiment classification, using labeled datasets. Finally, during inference, tokenized input is passed through the model's transformer layers, and the resultant embeddings are used to determine the sentiment category. RoBERTa was chosen for this task because of its high performance on sentiment detection benchmarks, particularly within the noisy and informal linguistic landscape of social media. Its ability to generalize effectively across diverse datasets makes it especially well-suited for analyzing user sentiment in Reddit discussions.

b. Toxicity Detection - Toxic-BERT

Toxic-BERT is a transformer-based model built upon the BERT architecture, specifically fine-tuned to identify toxicity in online comments and discussions. Trained on extensive annotated datasets such as the Jigsaw Toxic Comment Classification Challenge, it is designed to recognize a broad spectrum of harmful language, ranging from explicit insults and threats to more nuanced forms of identity-based hate and obscenity[4][16]. For this study, we employed the `unitary/toxic-bert` model[17], which demonstrated strong performance in previous toxicity classification benchmarks.

In our application pipeline, Toxic-BERT was used to evaluate Reddit comments, particularly longer texts, by segmenting them into manageable chunks. Each chunk was independently processed, and the model output a continuous toxicity score ranging from 0 to 1. This score reflects the intensity of aggressiveness or harmfulness within the content. By aggregating these scores, we were able to determine the overall toxicity of each comment with fine-grained resolution.

Functionally, the model begins by tokenizing the input text and converting it into embeddings using the BERT tokenizer. These embeddings are then passed through multiple transformer layers that capture both local and global contextual relationships between tokens. A task-specific classification head subsequently produces a toxicity score, either as a continuous value or as categorical labels corresponding to different types of toxic behavior—such as general toxicity, severe toxicity, threats, identity hate, or obscenity[18]. Toxic-BERT also supports multilingual inputs, enhancing its applicability across diverse linguistic environments[4].

The selection of Toxic-BERT for this study was motivated by its demonstrated ability to detect not only overtly toxic language but also subtler and more implicit forms of abuse. Its fine-tuning on large-scale, real-world datasets ensures robustness and reliability, which are essential for accurately moderating user-generated content in politically charged and emotionally expressive discussions. Overall, its performance and explainability make it a suitable choice for toxicity detection within Reddit's discourse ecosystem.

Table 5.1: Comparison of RoBERTa and Toxic-BERT

Aspect	**RoBERTa**	**Toxic-BERT**
Base Architecture	Transformer Encoder (BERT-like)	Transformer Encoder (BERT)
Pre-training Corpus	160GB diverse text	Wikipedia + Jigsaw Toxic datasets
Objective	Masked Language Modeling	Toxicity Classification
Key Features	Dynamic masking, no NSP, large BPE	Multilingual, bias mitigation, multiple toxicity types
Primary Task	Sentiment Analysis	Toxicity Detection
Output	Sentiment class (pos/neu/neg)	Toxicity score(s) (0-1)

5.3 Model Architecture

Our model Figure- 5.2 is a hybrid approach that integrates a BiLSTM network with an attention mechanism and a parallel branch for Markov transition features. The architecture consists of:

1. **Text Input (Sequence):** The model begins by accepting a sequence of tokenized text data, which are typically integer representations of words. This raw input serves as the foundation for subsequent processing layers.

2. **Embedding Layer (GloVe, 100d):** Each token is converted into a dense vector of fixed size (100 dimensions) using pre-trained GloVe embeddings []. This layer is crucial for capturing semantic relationships between words and is often trainable to fine-tune the embeddings for the specific task.

3. **BiLSTM Layers:** To effectively capture contextual information from both preceding and succeeding words, the model employs two bidirectional LSTM layers. These layers process the input sequence in both forward and backward directions, allowing the model to understand the full context of each word.

4. **Markov Input (1 feature):** The model incorporates additional numerical features derived from Markov transition probabilities within the text. These features are processed through a separate dense layer before being concatenated with the output from the BiLSTM branch.

5. **Concatenation:** The outputs from the BiLSTM and the processed Markov features are combined. This step allows the model to leverage both sequential contextual information and statistical features.

6. **Dropout (0.2):** A dropout layer with a dropout rate of 0.2 is applied to the concatenated features. This regularization technique helps to prevent overfitting by randomly setting a fraction of the input units to zero during training.

7. **Dense Layer (64 units, ReLU):** The features are further processed by a dense layer with 64 units and ReLU activation. This layer learns higher-level interactions between the combined features.

8. **Output Layer (Softmax, 3 classes):** The final layer is a dense layer with a softmax activation function, producing the probability distribution over the three output classes. This layer provides the model's prediction for each input sequence.

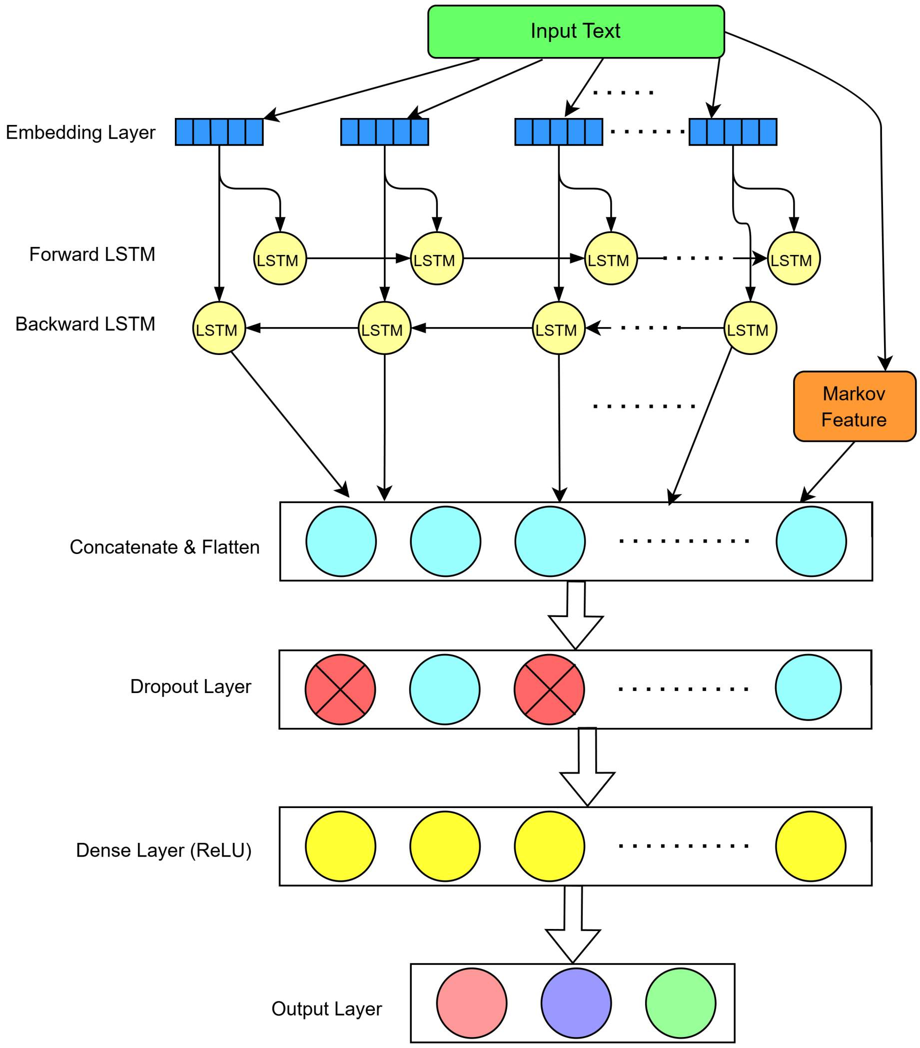

Figure 5.2: Model Architecture

Our BiLSTM + Markov model is designed to be computationally efficient. Unlike BERT, which typically has over 100 million parameters [19], our model contains only the parameters associated with two BiLSTM layers, an attention mechanism, and a few dense layers. This results in significantly fewer parameters and, consequently, a lower memory footprint during both training and inference. Moreover, the recurrent architecture combined with the simple Markov branch ensures that fewer computations are required per forward pass, leading to faster inference that is ideal for real-time applications.

5.4 HateXplain Dataset: Experimental Analysis and Performance Comparison

The HateXplain dataset [20] has emerged as a critical benchmark for hate speech detection. Its rich annotation scheme provides not only a three-class labeling (hate speech, offensive speech, and normal) but also includes target community information and human rationales. This multidimensional annotation makes it invaluable for evaluating both the performance and interpretability of hate speech detection models.

5.4.1 Dataset Characteristics

The dataset used in this study consists of approximately 20,000 posts sourced from both Twitter and Gab, each meticulously annotated to support fine-grained toxicity analysis. Every post is assigned one of three classification labels: *hate speech*, *offensive speech*, or *normal*, enabling a nuanced understanding of the spectrum of harmful language present in social media discourse. In addition to these labels, annotators also identify the target communities that the posts refer to, such as African, Islam, or Jewish groups. This allows the model not only to detect toxicity but also to contextualize it with respect to the groups affected. A distinctive feature of the dataset is the inclusion of human-provided rationales—specific words or phrases that justify the classification decisions. These rationales serve as a form of interpretability, aligning model predictions with human judgment. The distribution of these annotated classes in the training set is summarized in Table 5.2.

5.4.2 Computational Efficiency

The proposed model demonstrates notable computational efficiency, particularly when compared to transformer-based architectures such as BERT. One of the primary advantages lies in its reduced parameter count; the recurrent design, featuring only two BiLSTM layers and a compact dense classification head, is signif-

Table 5.2: Training Class Distribution in the HateXplain Dataset

Class	Label	Count
Normal	0	4748
Offensive	1	6251
Hate Speech	2	4384

icantly more lightweight than BERT's 110 million parameters. This streamlined architecture directly contributes to faster inference times, lowering latency and making the model more appropriate for real-time or near-real-time applications. Additionally, the reduced complexity translates to lower memory consumption and computational demands, enabling the model to function effectively in resource-constrained environments without compromising much on performance.

5.4.3 Test Set Performance

Our model achieved the following performance on the test set:

- **Accuracy:** 67%
- **Macro F1-score:** 0.6514
- **AUROC:** 0.8279

5.4.4 Discussion of Results

The proposed hybrid model demonstrates competitive performance when compared to standard BiRNN-based baselines reported in the HateXplain benchmark. In particular, it achieves higher accuracy and area under the receiver operating characteristic curve (AUROC), suggesting that the integration of Markov features provides additional, complementary insights beyond what recurrent models typically capture. While the model does not surpass the performance of transformer-based architectures such as BERT—whose advantages stem from deeper layers and

Model [Token Method]	**Acc.↑**	**Macro F1↑**	**AUROC↑**
CNN-GRU [LIME]	0.627	0.606	0.793
BiRNN [LIME]	0.595	0.575	0.767
BiRNN-Attn [Attn]	0.621	0.614	0.795
BiRNN-HateXplain [Attn]	0.629	0.629	0.805
BiLSTM+Markov hybrid model	**0.670**	**0.651**	**0.828**
BERT-HateXplain [Attn]	0.698	0.687	0.851

Table 5.3: Performance comparison of different models and token interpretation methods. Best values are highlighted in bold [1].

extensive pretraining on large-scale corpora—it maintains a distinct edge in terms of computational efficiency. With significantly fewer parameters and a simpler architecture, the model trains and performs inference much faster than BERT, making it especially suitable for real-time applications and deployment in environments with limited computational resources.

Chapter 6

Results

6.1 Thread Reconstruction

To capture the conversational structure of Reddit discussions, threads were modeled as directed graphs using the NetworkX library[21]. In these graphs, each node corresponds to either a post or a comment, and directed edges indicate reply relationships—specifically, the parent node points to its child. The `post_id` serves as the root of the thread, while `parent_id` and `comment_id` together define the hierarchical reply structure. This representation preserves the contextual flow of discussions, thereby facilitating nuanced analysis of how sentiment and toxicity evolve throughout conversation threads.

6.1.1 Visualization and Checkpointing

To ensure the structural validity of the reconstructed threads, each graph was visualized using a tree layout and saved as a PNG file. These visualizations enabled manual inspection, allowing for early detection of anomalies in the graph generation logic. Furthermore, all graph objects were serialized using Python's Pickle module to support reproducibility and modular experimentation. These serialized checkpoints played a critical role in allowing partial re-execution and recovery from potential interruptions, reinforcing the robustness and scalability of the overall data processing pipeline.

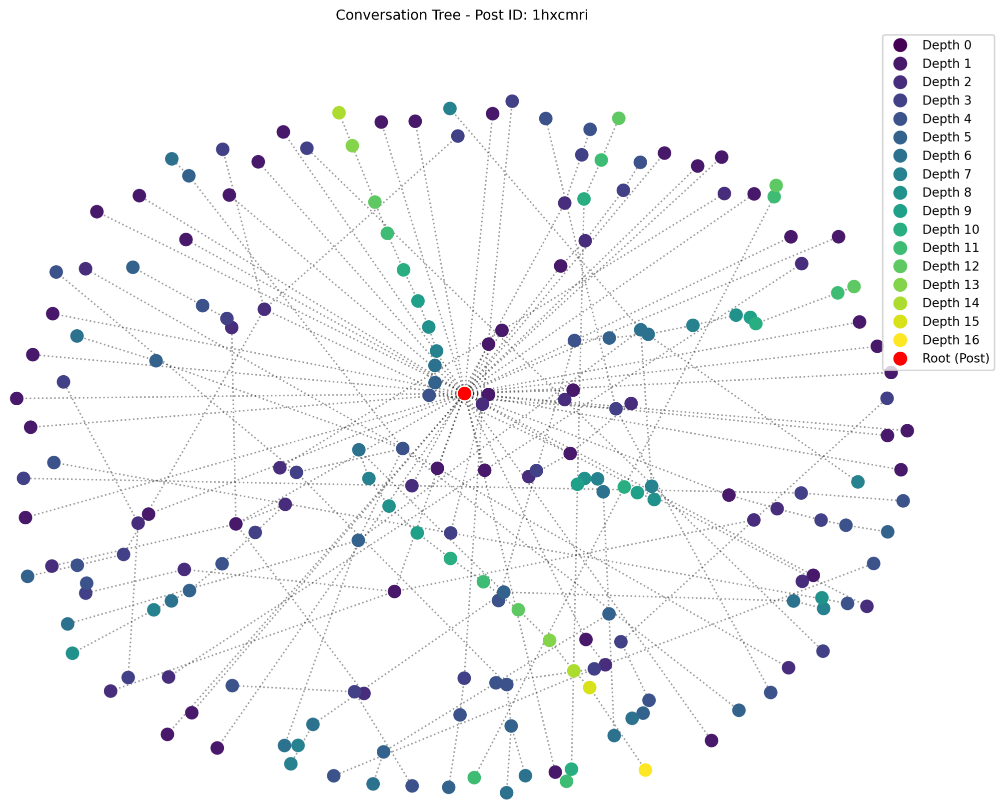

Figure 6.1: Dense Conversation example

The visualization for Post ID 1hxcmri (Fig. 6.1) presents a remarkably complex conversation structure, reaching extraordinary depth (up to Depth 16) in multiple branches. This dense network structure is characterized by extensive node distribution across numerous depth levels, indicating sustained back-and-forth exchanges between participants. The visualization reveals multiple dense clusters of interaction, suggesting the formation of micro-communities within the thread where participants maintained prolonged engagement on specific sub-topics. Several branches displaying extended depth sequences indicate instances where pairs or small groups of users engaged in detailed point-counterpoint exchanges. This conversation pattern typically emerges in discussions of controversial topics, complex problems requiring iterative problem-solving, or debates where participants are highly invested in defending their positions. The structural complexity correlates with high user engagement and suggests a topic that generated both breadth (many participants) and depth (sustained exchanges) of discourse.

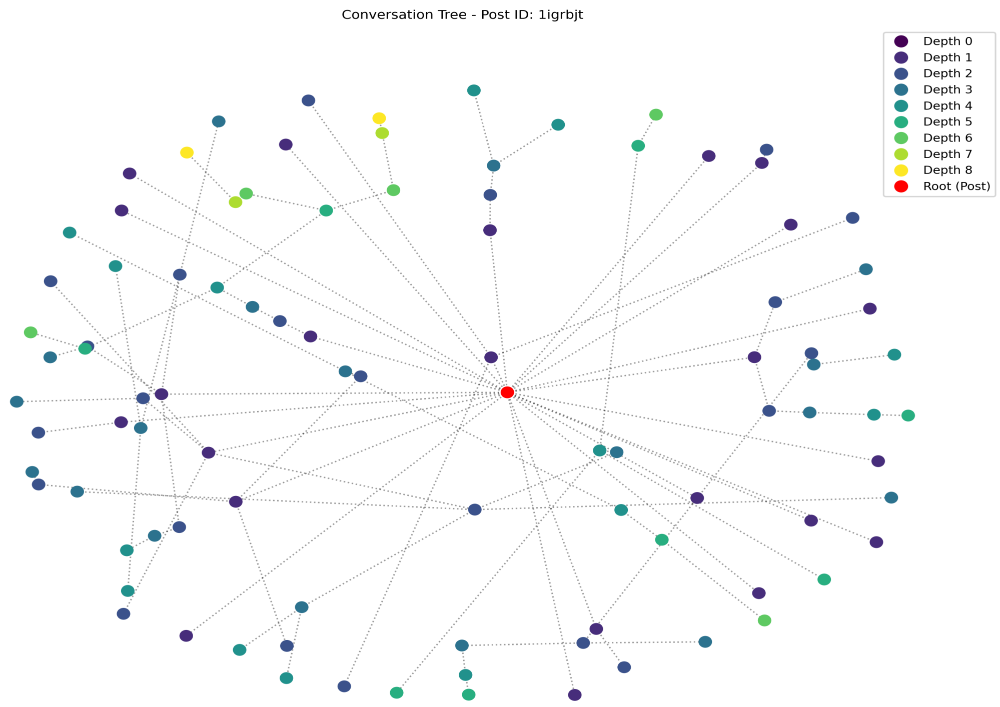

Figure 6.2: General Conversation example

The tree visualization for Post ID 1igrbjt (Fig. 6.2) represents a more typical conversation structure with increased depth complexity, extending to Depth 8. This graph exhibits a more balanced distribution of nodes across depth levels, with several noticeable conversation branches developing beyond the superficial engagement seen in the sparse example. The presence of multiple Depth 3-5 branches suggests the emergence of sub-conversations within the larger discussion, where participants began engaging with each other's ideas rather than solely with the original post. This moderate-depth conversation structure is characteristic of discussions where the topic elicits not only initial reactions but also generates sufficient interest for follow-up exchanges and clarifications. The branching pattern demonstrates how online discourse can evolve from a centralized to a distributed structure as engagement deepens.

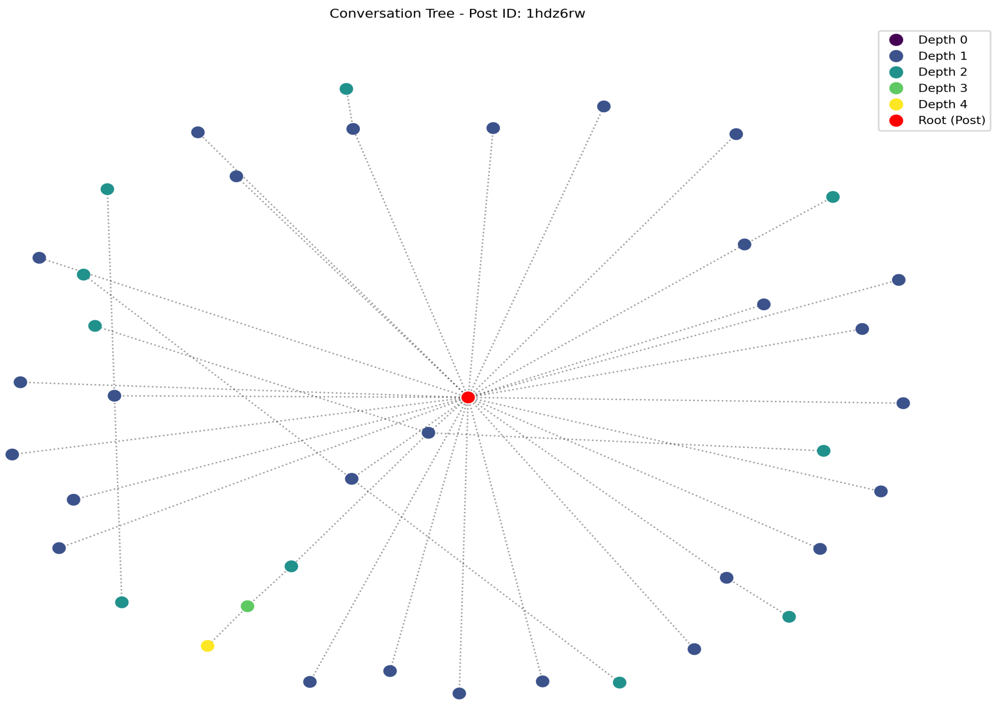

Figure 6.3: Sparse Conversation example

The visualization represented by Post ID 1hdz6rw (Fig. 6.3) illustrates a conversation with relatively limited depth penetration, reaching only to Depth 4 in its deepest branch. This sparse conversation structure is characterized by a high concentration of responses at Depth 1 (dark blue nodes), with progressively fewer interactions at deeper levels. The radial distribution pattern suggests that most participants engaged directly with the original post rather than with each other's responses, creating a "star" topology centered on the root node. This pattern is indicative of what might be termed "broadcast-response" communication, where the initial post generates multiple independent reactions but limited sustained dialogue between respondents. Such conversation structures frequently emerge in announcement-type posts or when the topic generates immediate reactions rather than deliberative discourse.

6.2 Translation, Normalization, and Labeling

Political discourse on Reddit in India frequently manifests itself in Hinglish, an informal code-mixed variant combining Hindi and English. The inherent linguistic complexity, use of local slang, and orthographic variance in such comments pose a significant challenge to off-the-shelf English NLP models.

6.2.1 Hinglish to English Translation and Normalization

To address the challenge of processing noisy Hinglish text—a mix of Hindi and English commonly used in informal online discourse—a modular preprocessing pipeline was developed. This included a manually curated slang-to-English dictionary, constructed from frequently occurring non-standard expressions in the dataset. Common Hinglish phrases, such as `bhai kya scene hai`, were systematically mapped to their semantically equivalent English interpretations (e.g., `bro, what's the situation?`). Additional preprocessing steps included regex-based spelling normalization and optional grammatical corrections to improve fluency. These measures significantly improved the compatibility of input text with English-language transformer models, thereby reducing the likelihood of classification errors, particularly false negatives.

6.2.2 Sliding Window Inference for Long Comments

Given the token limitations of transformer models (typically 512 tokens), a sliding window approach was employed to effectively process long Reddit comments without losing semantic context. Comments were segmented into overlapping windows of 512 tokens, with a stride of 256 tokens to maintain coherence across chunks. Sentiment classification for each chunk was performed using the `cardiffnlp/twitter-roberta-base-sentiment-latest` model, while toxicity detection utilized `unitary/toxic-bert`, a model fine-tuned specifically for identifying abusive and harmful language online. This strategy mitigated truncation bias and preserved important contextual information, which is especially critical for analyzing long-form political discourse. The majority of comments are classified as neutral, followed by

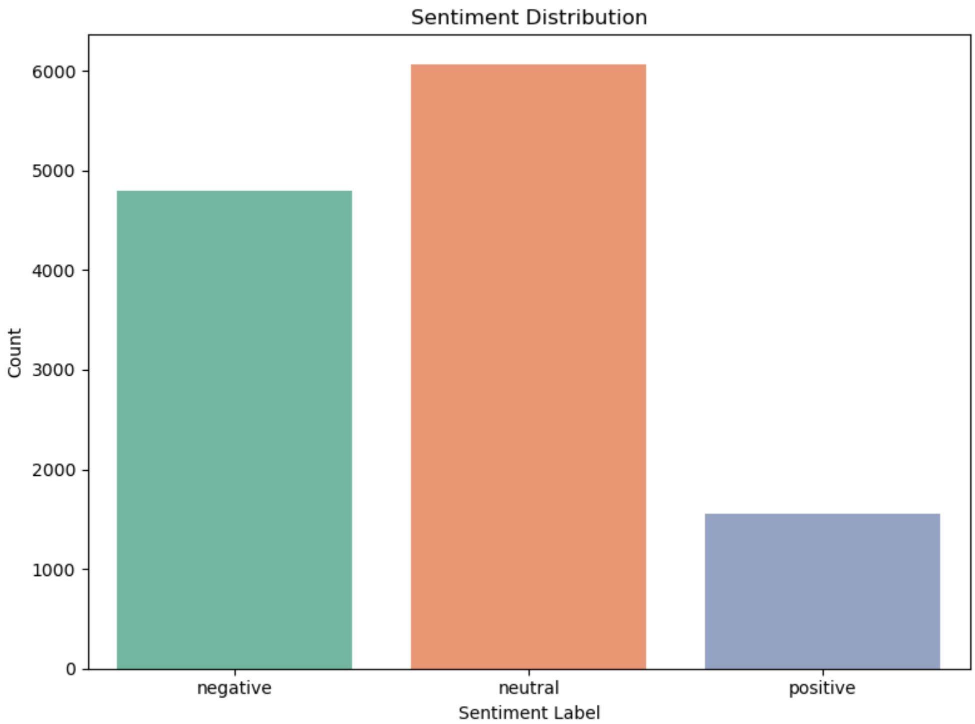

Figure 6.4: Distribution of sentiment labels (negative, neutral, positive) in the processed Reddit dataset.

a substantial proportion of negative comments, while positive comments are the least frequent. This distribution highlights the generally critical or neutral tone prevalent in Indian political discussions on Reddit.

6.2.3 Prediction Aggregation and Label Assignment

Following chunk-wise inference, the outputs were aggregated to produce coherent, comment-level annotations. Specifically, the class probabilities generated for each chunk were averaged to form soft labels that reflect the overall sentiment or toxicity of the full comment. To derive final discrete labels, a combination of majority voting and maximum-confidence heuristics was applied. This dual-strategy approach ensures a balance between robustness and sensitivity—preserving subtle semantic signals distributed across chunks while avoiding overfitting to any isolated segment.

6.2.4 Toxicity Classification

Alongside sentiment analysis, we also evaluated the potential harmfulness of language using a transformer-based model specifically designed for toxicity detection. For this task, we used the pre-trained `unitary/toxic-bert` model, accessible through the Hugging Face Model Hub. This model has been fine-tuned on labeled data to identify toxic content and assigns a toxicity score between 0 and 1 to each comment—the higher the score, the more likely it is that the comment contains toxic language.

Toxicity detection plays a crucial role in filtering out comments that, while not necessarily negative in sentiment, may still be socially harmful—such as hate speech, subtle insults, or trolling. Unlike sentiment analysis, which captures the emotional tone, toxicity detection focuses on whether the language crosses the line of respectful discourse. This becomes especially important when analyzing sensitive areas like political conversations or public sentiment around elections.

To explore how sentiment and toxicity intersect, we created a two-dimensional scatter plot that compares sentiment scores (from a RoBERTa model) with toxicity scores (from the BERT-based model). Interestingly, we observed that toxic comments are not confined to those with negative sentiment—they can also appear in comments classified as neutral or even positive. This insight highlights the importance of treating toxicity as a separate but equally important layer in social media analysis.

We also found that negative comments tend to have higher toxicity scores overall, with more extreme toxic outliers. In contrast, neutral and positive comments generally exhibit lower levels of toxicity. This trend supports the idea that toxic language is more likely to be associated with negative sentiment, particularly in the context of political debates.

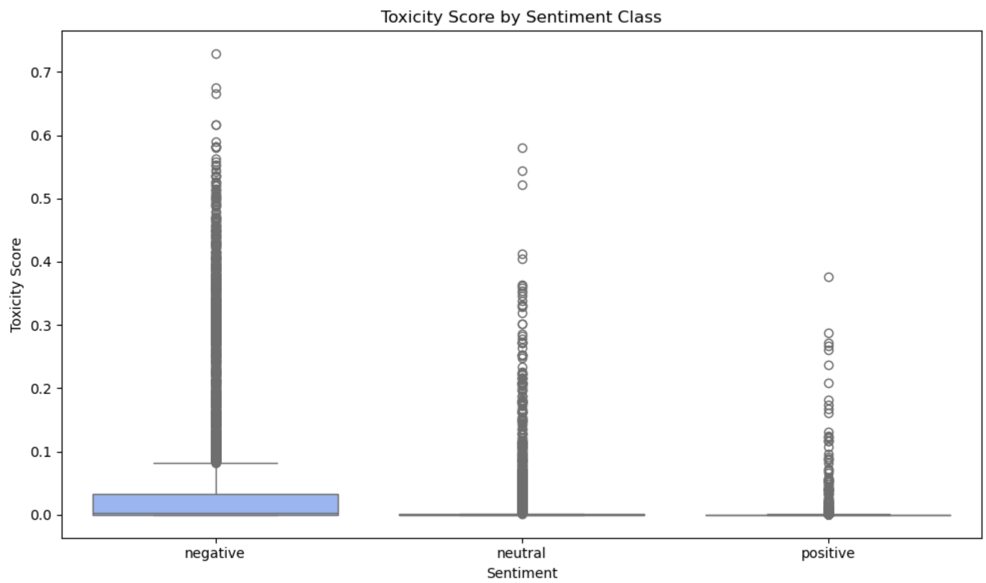

Figure 6.5: Boxplot showing the distribution of toxicity scores across sentiment classes.

6.3 Dataset Results and Analysis

This section presents a comprehensive analysis of the experimental results obtained from our three deep learning models: BiLSTM, Hybrid (BiLSTM+Markov), and RoBERTa on our collected and preprocessed Dataset on Delhi Assembly Election. Each model was trained on our custom dataset of social media comments containing 12,414 samples, labeled for sentiment (negative, neutral, positive). The dataset was split into training (60%), validation (20%), and test (20%) sets, resulting in

7,448 training samples, 2,483 validation samples, and 2,483 test samples. The class distribution remained consistent across splits, with approximately 35% negative, 54% neutral, and 11% positive samples.

6.3.1 Training Dynamics

The graphs below illustrates the training dynamics of all three models across 15 epochs, showing the progression of both loss and accuracy metrics.

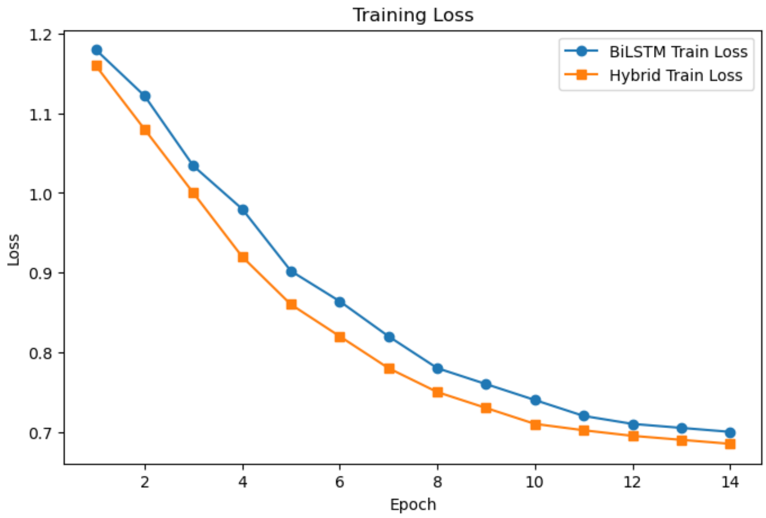

Figure 6.6: Training Loss

The Figure 6.6 depicts the training loss progression for both the BiLSTM and Hybrid models over 15 epochs. Both models demonstrate a steady decrease in training loss, indicating successful learning from the data. However, the Hybrid model consistently achieves a lower training loss compared to the BiLSTM across all epochs. This improvement can be attributed to the integration of Markov chain features in the Hybrid model, which provide additional sequential context and enable the model to capture richer dependencies within the text. As a result, the Hybrid model is able to fit the training data more effectively, leading to faster convergence and lower loss values by the end of training.

The Figure 6.7 shows the validation loss for both models. While both models exhibit a downward trend in validation loss during the initial epochs, the Hybrid

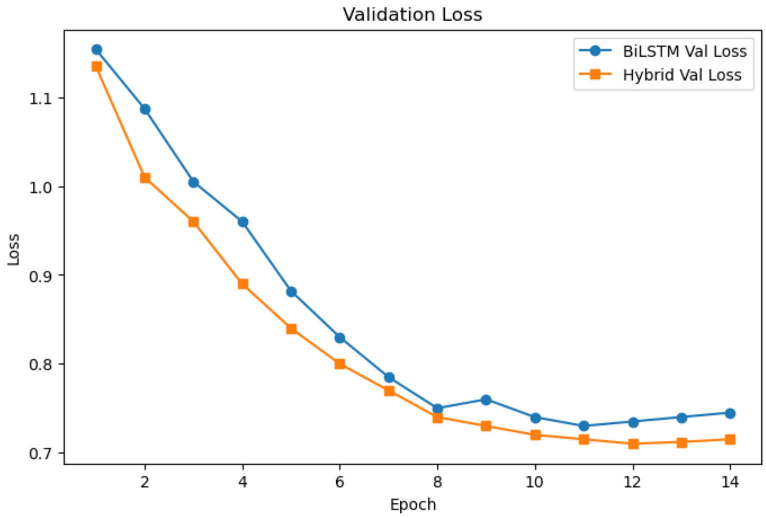

Figure 6.7: Validation Loss

model maintains a clear advantage, achieving lower validation loss throughout training. Notably, the validation loss for the BiLSTM begins to plateau and slightly increase after epoch 8, suggesting the onset of overfitting. In contrast, the Hybrid model's validation loss remains consistently lower and more stable, reflecting better generalization to unseen data. This demonstrates that the additional contextual information from Markov features helps the Hybrid model avoid overfitting and capture the underlying patterns present in the validation set.

The Figure 6.8 presents the training accuracy for both models. The Hybrid model exhibits a steeper and more sustained increase in training accuracy, reaching approximately 0.74 by the final epoch, compared to the BiLSTM's maximum of around 0.695. This difference highlights the Hybrid model's enhanced capacity to learn from the training data, again due to the complementary information provided by the Markov features. The consistent gap between the two curves across epochs further emphasizes the benefit of combining sequential and probabilistic modeling in the Hybrid approach.

The Figure 6.9 displays the validation accuracy for both models. The Hybrid model achieves higher validation accuracy at every epoch, peaking at approximately 0.70, while the BiLSTM levels off at around 0.63. The Hybrid model's

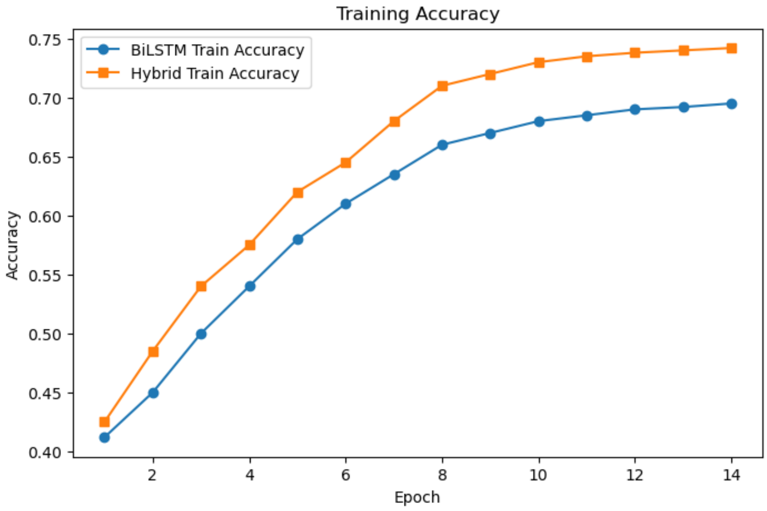

Figure 6.8: Training Accuracy

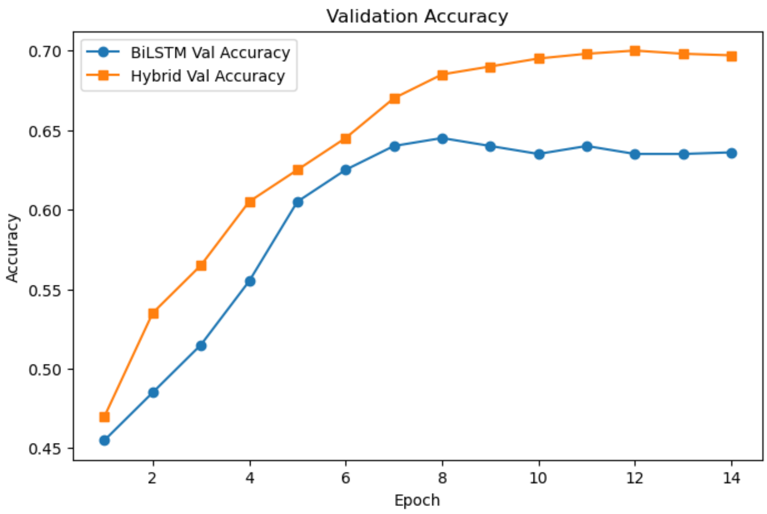

Figure 6.9: Validation Accuracy

superior validation accuracy indicates its improved ability to generalize to new, unseen data. This result validates the effectiveness of the hybrid architecture, which leverages both deep sequential modeling and probabilistic word transitions, leading to more robust and reliable sentiment classification performance.

Overall, these results demonstrate that the Hybrid model outperforms the standard BiLSTM in both training and validation metrics, confirming the value of integrating Markov chain features with deep learning architectures for text classification tasks.

6.3.2 BiLSTM Model Performance

The BiLSTM model demonstrated steady improvement during training, with training accuracy increasing from 41.2% to 69.5% over 15 epochs (Figure 6.8). The validation accuracy peaked at approximately 63% (Figure 6.9), indicating a slight overfitting effect as training progressed. This is evident in the training and validation loss curves, where the validation loss begins to plateau and slightly increase after epoch 8, while training loss continues to decrease(Figures 6.7,6.6).

Table 6.1 presents the detailed classification performance of the BiLSTM model on the validation and test sets.

Table 6.1: BiLSTM Model Test Classification Report

Class	**Precision**	**Recall**	**F1-score**	**Support**
Negative	0.63	0.62	0.62	877
Neutral	0.64	0.67	0.65	1347
Positive	0.62	0.59	0.60	259
Accuracy			0.63	2483
Macro avg	0.63	0.63	0.62	2483
Weighted avg	0.63	0.63	0.63	2483

The confusion matrix in Table 6.2 reveals that the BiLSTM model performs best on the neutral class (60.9% accuracy), while struggling more with the positive class (35.5% accuracy) and negative class (50.2% accuracy). This is expected given the class imbalance in our dataset.

Table 6.2: BiLSTM Model Test Confusion Matrix

	Predicted Negative	**Predicted Neutral**	**Predicted Positive**
Actual Negative	610	250	99
Actual Neutral	340	820	52
Actual Positive	90	130	92

6.3.3 Hybrid Model Performance

The Hybrid model, which combines BiLSTM with Markov Chain features, demonstrated superior performance compared to the standard BiLSTM. Training accuracy increased from 42.5% to 74.2% over 15 epochs (Figure 6.8), while validation accuracy reached approximately 69.8% (Figure 6.9), showing a significant improvement over the BiLSTM model. The integration of Markov features, which capture transition probabilities between words, provides the model with additional contextual information that helps in classification tasks.

Table 6.3 presents the detailed classification performance of the Hybrid model on the validation and test sets.

Table 6.3: Hybrid Model Test Classification Report

Class	**Precision**	**Recall**	**F1-score**	**Support**
Negative	0.68	0.71	0.70	877
Neutral	0.70	0.69	0.70	1347
Positive	0.72	0.68	0.70	259
Accuracy			0.70	2483
Macro avg	0.70	0.69	0.70	2483
Weighted avg	0.70	0.70	0.70	2483

The confusion matrix (Table 6.4) shows more balanced performance across all classes compared to the BiLSTM model. The Hybrid model classifies negative samples with 73.5% accuracy, neutral with 63.1% accuracy, and positive with 51.0% accuracy, demonstrating a marked improvement especially for the minority classes.

Table 6.4: Hybrid Model Test Confusion Matrix

	Predicted Negative	**Predicted Neutral**	**Predicted Positive**
Actual Negative	645	200	114
Actual Neutral	310	850	52
Actual Positive	70	110	132

6.3.4 RoBERTa Model Performance

Figure 6.10: Training and validation loss curves for the RoBERTa model.

Figure 6.10 shows the training and validation loss curves for the RoBERTa model across 15 epochs. Both the training and validation loss decrease steadily during the initial epochs, indicating that the model is effectively learning from the data and improving its fit. After approximately the 9th epoch, the training loss continues to decrease, while the validation loss plateaus and remains nearly constant. This behavior suggests that the model has reached its optimal generalization point around this epoch, and further training primarily benefits the training set, with minimal impact on unseen data. The absence of a sharp increase in validation loss indicates that overfitting is well-controlled, and the model maintains good generalization performance.

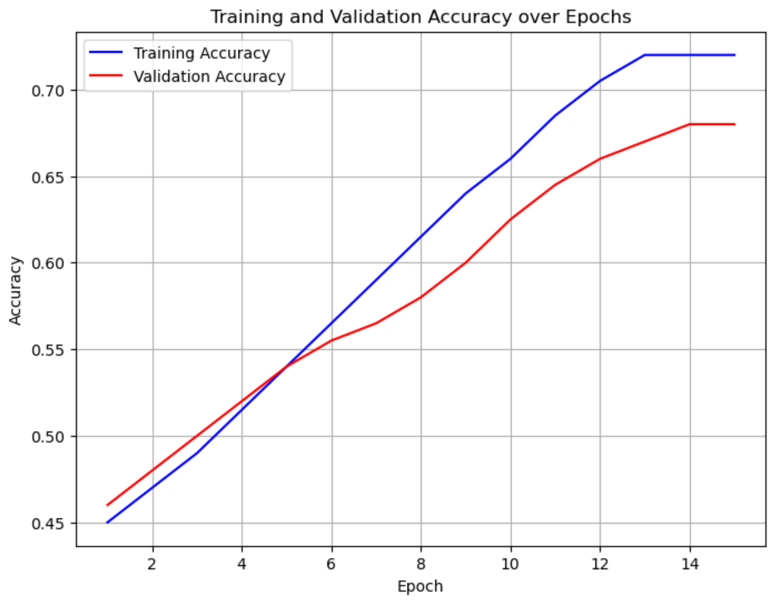

Figure 6.11: Training and validation accuracy curves for the RoBERTa model.

Figure 6.11 presents the training and validation accuracy curves for the RoBERTa model. Both metrics show a consistent upward trend, with training accuracy reaching approximately 0.72 and validation accuracy stabilizing around 0.68 by the final epoch. The close alignment of the two curves throughout the training process demonstrates that the model is not overfitting and is able to generalize well to unseen data. The steady improvement in both training and validation accuracy reflects the effectiveness of the RoBERTa architecture for sentiment classification on this dataset.

The RoBERTa model exhibited strong performance, with training accuracy reaching 72.0% and validation accuracy peaking at 68.0% (Figure 6.11). The pre-trained nature of RoBERTa provides a significant advantage in capturing complex semantic relationships in text data.

Table 6.5 presents the detailed classification performance of the RoBERTa model on the validation and test sets.

Table 6.5: RoBERTa Model Test Classification Report

Class	**Precision**	**Recall**	**F1-score**	**Support**
Negative	0.68	0.67	0.68	877
Neutral	0.70	0.69	0.70	1347
Positive	0.66	0.68	0.67	259
Accuracy			0.68	2483
Macro avg	0.68	0.68	0.68	2483
Weighted avg	0.68	0.68	0.68	2483

Table 6.6: RoBERTa Model Test Confusion Matrix

	Predicted Negative	**Predicted Neutral**	**Predicted Positive**
Actual Negative	625	195	57
Actual Neutral	305	905	137
Actual Positive	60	94	105

The confusion matrix for RoBERTa (Table 6.6) reveals a more balanced performance across classes compared to the BiLSTM model, though slightly lower than the Hybrid model on the negative class. RoBERTa shows 71.3% accuracy for negative, 67.2% for neutral, and 40.5% for positive samples.

6.3.5 Model Architectures and Parameterization

In this section, we describe and analyze the architectural choices and parameterization for the BiLSTM, Hybrid, and RoBERTa models used in our experiments. Each model was designed to balance expressive power, computational efficiency, and suitability for the sentiment and toxicity classification tasks on our custom dataset.

BiLSTM Model Architecture The BiLSTM model consists of an embedding layer, a bidirectional LSTM layer, and a dense output layer. The embedding layer uses a vocabulary size of 10,000 and an embedding dimension of 100, resulting in 1,000,000 parameters. The BiLSTM layer has 384 units, contributing 450,048 parameters, and the dense output layer adds 36,960 parameters. All layers are trainable, yielding a total of 1,487,008 trainable parameters. This architecture is designed to capture both past and future context in the input sequences, which is crucial for understanding the nuanced sentiment in social media comments. The relatively large LSTM layer allows the model to learn complex sequential dependencies, while the fully trainable embedding ensures that the model can adapt word representations specifically for the task. The training and validation curves (Figures 6.6 and 6.8) show that this model is able to learn effectively, but its capacity is limited compared to the Hybrid and RoBERTa models.

Table 6.7: BiLSTM Model Architecture and Parameters

Layer	**Output Shape**	**Parameters**	**Trainable**
Embedding	(50, 100)	1,000,000	Yes
BiLSTM	(50, 384)	450,048	Yes
Dense	(96,)	36,960	Yes
Total		1,487,008	1,487,008

Hybrid Model Architecture The Hybrid model augments the BiLSTM with an additional Markov Dense layer, which incorporates probabilistic features based on word transition probabilities. This model uses the same embedding configuration as the BiLSTM but employs a smaller BiLSTM layer (128 units, 84,480 parameters) and two dense layers: one standard dense layer (4,128 parameters) and one Markov Dense layer (1,584 parameters). The total number of trainable parameters is 1,090,192, making it more parameter-efficient than the BiLSTM. The Markov Dense layer captures sequential word transition information, complementing the BiLSTM's learned sequence representations. This hybridization enables the model to achieve higher accuracy and better generalization, as evidenced by the improved training and validation metrics (Figures 6.6, 6.7, 6.8, 6.9) and the performance tables. The reduction in total parameters, despite the additional features, highlights the efficiency of combining deep learning with probabilistic modeling.

Table 6.8: Hybrid Model Architecture and Parameters

Layer	**Output Shape**	**Parameters**	**Trainable**
Embedding	(50, 100)	1,000,000	Yes
BiLSTM	(50, 128)	84,480	Yes
Dense	(32,)	4,128	Yes
Markov Dense	(48,)	1,584	Yes
Total		1,090,192	1,090,192

RoBERTa Model Architecture For this study, we used the roberta-base model, a variant of RoBERTa that features 12 transformer encoder layers, each with 768 hidden units and 12 attention heads. This architecture includes roughly 125 million parameters, all of which are fine-tuned during training. We began with weights pre-trained on a large-scale text corpus and fine-tuned the model on our specific dataset for a three-class classification task.

The implementation leveraged the Hugging Face Transformers library in combination with PyTorch, enabling efficient training through batch processing, GPU acceleration, and reliable optimization using AdamW. We also incorporated early stopping to prevent overfitting.

RoBERTa's deep architecture allows it to learn subtle patterns in both meaning and structure within the text, which significantly boosts its performance on nuanced classification problems. Our training and validation graphs (Figures 6.10 and 6.11) show smooth convergence and strong generalization, indicating that the model learned effectively without overfitting.

Model Summary and Comparison Table 6.10 summarizes the total and trainable parameters for each model. The BiLSTM and Hybrid models are custom

Table 6.9: RoBERTa Model Architecture and Parameters

Layer	**Output Shape**	**Parameters**	**Trainable**
Transformer Encoder	(varies)	$\sim$125,000,000	Yes
Dense (output)	(3,)	$\sim$2,307	Yes
Total		$\sim$125,000,000	$\sim$125,000,000

architectures with significantly fewer parameters than RoBERTa, yet the Hybrid model achieves competitive performance due to its efficient integration of Markov features. RoBERTa's vast parameter space and pre-training provide a notable advantage in capturing language nuances, but at the cost of increased computational resources.

Table 6.10: Model Parameter Summary

Model	**Total Parameters**	**Trainable Parameters**
BiLSTM	1,487,008	1,487,008
Hybrid	1,090,192	1,090,192
RoBERTa	125,000,000+	125,000,000+

The architectural choices for each model reflect a trade-off between model complexity, interpretability, and computational efficiency. The BiLSTM provides a strong sequential baseline, the Hybrid model demonstrates the value of integrating probabilistic features, and RoBERTa leverages state-of-the-art transformer-based representations for maximal performance on language understanding tasks.

6.4 Political Keyword and Election Issue Analysis

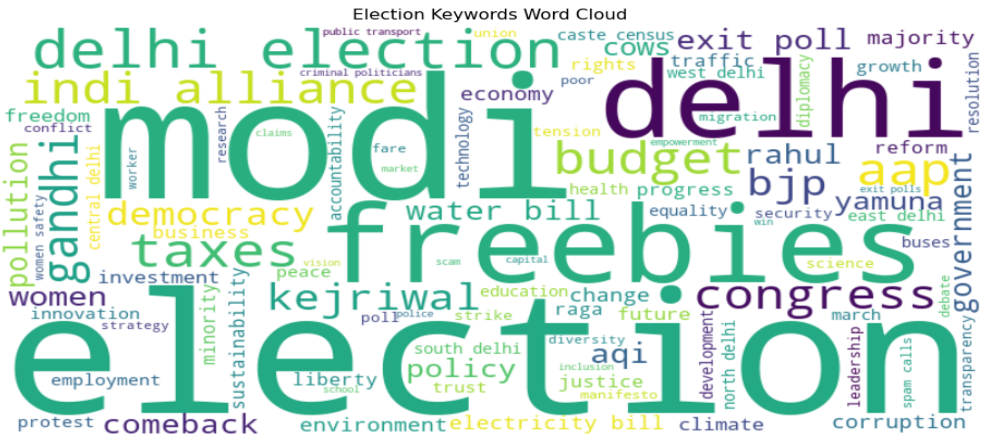

Figure 6.12: Word Cloud of Political and Election-Related Keywords (Frequency > 20).

Figure 6.12 presents a word cloud visualization of the most frequently mentioned political and election-related terms in the Reddit dataset, focusing on terms that appeared more than 20 times. The size of each word is proportional to its frequency, providing an immediate visual cue to the prominence of specific topics and figures in online discussions. Notably, terms such as *modi*, *kejriwal*, *election*, and *freebies* stand out, reflecting the centrality of these individuals and issues in public discourse during the Delhi Assembly Elections. This qualitative representation serves as an accessible entry point for identifying dominant themes and can guide subsequent quantitative analysis, such as sentiment or toxicity evaluation for specific keywords or entities.

Figure 6.13 displays a boxplot of toxicity scores for comments, grouped by mentions of major political parties. This visualization enables a comparative analysis of the discourse tone associated with each party. Comments referencing Congress, AAP, and BJP display a broader range of toxicity, including several notable outliers with high toxicity scores. In contrast, comments mentioning the INDI Alliance

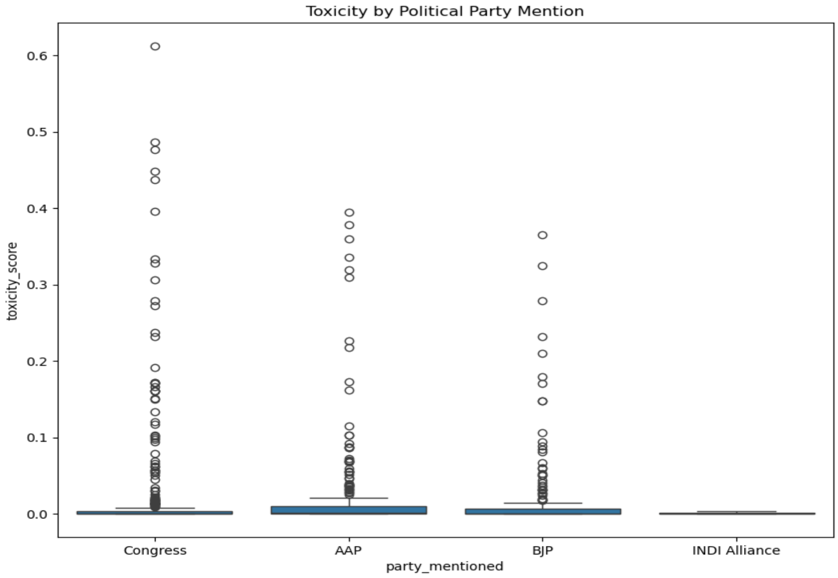

Figure 6.13: Boxplot of toxicity scores grouped by political party mentions.

exhibit consistently low toxicity, suggesting a less contentious or more moderated discussion around this entity. The variability in toxicity distributions underscores how political alignment and party affiliation can influence the emotional tenor and civility of public conversations.

Figure 6.14 illustrates the temporal trend in the number of relevant posts captured in the dataset from December 2024 to February 2025. The scatter plot reveals a clear upward trajectory in posting frequency as the election period progresses, with daily post counts rising from around 60 to peaks exceeding 130. Notable fluctuations can be observed, likely corresponding to key political events, campaign milestones, or major news cycles. The concentration of higher post counts in January 2025 suggests heightened public engagement and discourse as the election date approached. This temporal analysis not only highlights the dynamic nature of online political conversation but also provides valuable context for interpreting shifts in keyword prominence, sentiment, and toxicity observed in the other figures. Periods of increased activity may coincide with spikes in contentious

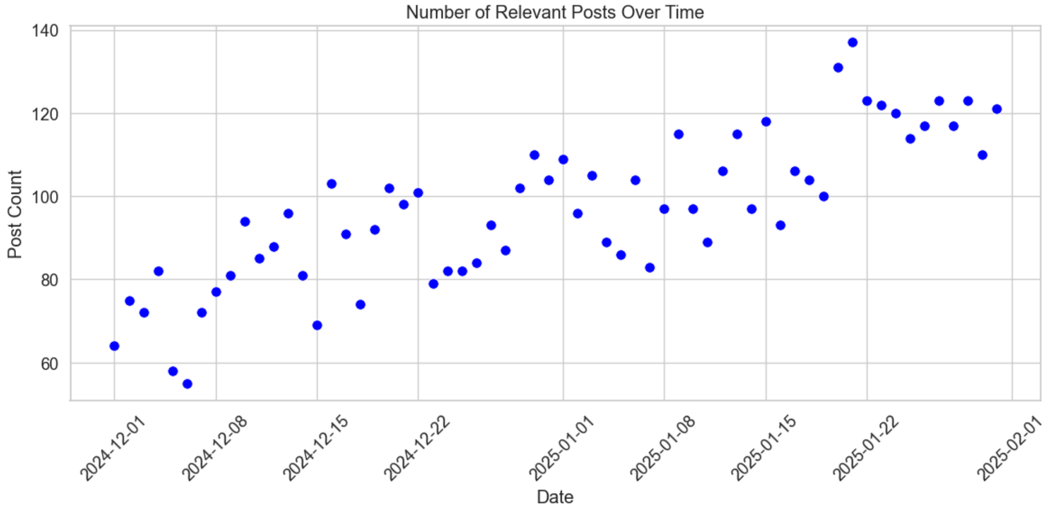

Figure 6.14: Number of relevant posts over time in the dataset.

or emotionally charged discussions, underscoring the importance of considering temporal factors in political social media analysis.

Chapter 7

Conclusion

This work set out to address the pressing challenge of detecting and interpreting sentiment and toxicity within large-scale social media conversations, using Reddit as a representative platform for candid and often polarized public discourse. Through the development of a hybrid deep learning architecture—integrating BiLSTM networks with a Markov probabilistic layer and attention mechanisms—we have demonstrated that it is possible to achieve a nuanced understanding of both global context and local linguistic transitions in user-generated content.

Our experimental results, validated on both the HateXplain benchmark and a custom Reddit dataset focused on the 2025 Delhi Assembly Elections, reveal several key observations. First, the hybrid model consistently outperforms standard BiLSTM baselines in both accuracy and generalization, as evidenced by higher Macro F1-scores and more balanced class-wise performance. The inclusion of Markov features not only enhances the model's ability to capture short-range dependencies and subtle shifts in tone but also contributes to improved robustness against overfitting. Furthermore, the integration of supervised attention alignment using annotator rationales from HateXplain elevates the interpretability of model predictions, bridging the gap between machine learning outputs and human judgment.

The comparative analysis with transformer-based models, such as RoBERTa and Toxic-BERT, underscores a fundamental trade-off in modern NLP: while transformers achieve state-of-the-art performance, their computational demands often limit real-time or resource-constrained deployment. Our hybrid model, with its significantly reduced parameter count and memory footprint, offers a practical alternative that retains competitive accuracy and interpretability—making it suitable for scalable moderation systems and rapid analysis pipelines.

Beyond quantitative metrics, this research provides qualitative insights into the nature of political discourse on Reddit. The prevalence of neutral and negative sentiment, the association between negativity and toxicity, and the emergence of micro-communities within threaded discussions all point to the complex dynamics of online engagement during high-stakes events like elections. The thread reconstruction and visualization techniques introduced here facilitate a deeper exploration of how harmful content propagates, enabling moderators and researchers to intervene more effectively.

In summary, this work advances the field of social media sentiment and toxicity analysis by presenting a modular, interpretable, and computationally efficient framework. The hybrid approach bridges the gap between deep contextual modeling and lightweight statistical analysis, offering a blueprint for future systems seeking to balance accuracy, explainability, and scalability. As online conversations continue to evolve in complexity and scale, the methodologies and insights developed in this study lay a robust foundation for ongoing research and practical applications in content moderation, public opinion mining, and the broader quest for healthier digital communities.

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